OPTIMAL PARAMETERIZATION FOR THE KNEED PASSIVE DYNAMIC BIPED ROBOT

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ABSTRACT
The study of dynamic bipedal robots has shown some configurations which are able to walk with a periodic gait on a slope with an appropriate angle, without any actuation. And the characteristics of the gait cycle are closely associated with the geometry and dynamics of the system.
This paper aims at designing a passive bipedal robot as an optimization problem in order to obtain the parameters which produce a periodic and symmetric gait. Additionally, it has been required that the robot to move a load specified by the user. The sought parameters are the size of the system, the location and the measurement of the mass center. And the optimization problem is carried out using the Differential Evolution Algorithm. Finally, a comparison between no optimal and optimal results are presented, via some simulations.

KEY WORDS
Dynamic bipedal walking, Limit Cycle, Parametric Optimization, Differential Evolution

1 Introduction

Presently, most of the systems made by the human, that require moving on a surface use the wheel as their principal element of movement. However, nature has produced a vast number of examples which use some sort of legs. By and large, they present many advantages over the wheel. Taking the example of the centipede with hundreds of legs, spiders with eight, insects with six, terrestrial mammals with four and the complex system of a biped. To get an insight into the complexity of the system skeletal action of humans, is sufficient to remember that its motion is performed by approximately 760 muscles [1]. Despite the fact that the model has been reduced to a reasonable number of degrees of freedom, there are still some complex phenomenons, represented by a hybrid system, which contains ordinary differential equations and algebraic transitions.

One definitive advantage of dynamic walking is the total energy conservation at low and moderated velocities. Therefore, the gravity effect is used for minimizing the muscle work [2]. Moreover, some works have shown that it is possible to walk in a completely passive way, when the biped robot moves along a slope, propelled only by gravity. This attribute is the principal property of a Passive Biped Robot (PBR) [3], [4], [5]. This is, probably, the principal reason that brings to light the importance of finding the optimal parameters in order to produce a natural and efficient walking. Also, the process of optimization, gives the opportunity to establish a set of functions to be able of quantifying the system performance and it imposes some constraints on the behavior of the biped.

Several optimization algorithms have been used to solve engineering design problems. A first approach to address this problem is the use of mathematical optimization techniques such as sequential quadratic programming (SQP). Nevertheless they need that the objective functions and constraints are twice continuously differentiable to calculate their gradient and Hessian [6]. However, for the particular system discussed in this paper, this attribute is not available and the traditional process is complicated enough to be applied

On the other hand, due to the growth of computing power that has been experienced since the last decades of the last century, the use of heuristic techniques for problem solving in engineering design has increased. These techniques are divided into two main groups, Evolutionary Algorithms (EA) and Collective Intelligence (CI) [7], which have been recurrently used because they produce good results as well as by computational simplicity of implementation.

The mathematical model of the kneed biped robot system has been broadly studied by several research groups [8], [9], [10], [11]. But in these works, the set of system parameters have been chosen without any particular process, and these values do not change in simulations, even in experiments. This method has been used because these works are particularly aimed at developing a robust control law. Where the controller is expected to solve many of the problems caused by poor configurations or uncertainties that are directly related to the system parameters.

In this paper, a single-objective dynamic optimization is proposed for solving the design problem of a knee passive dynamic biped robot. The optimization problem took into account the objective function and desired behavior, restrictions on system sizing and additional restrictions to move a load of 15 kilograms along a four degree slope, thus fulfilling a particular job.

This work is organized as follows: Section 2 presents the dynamic model and transition equations governing the behavior of the system. Optimization problem statement is presented in section 3, exposing the different elements that make up this problem: design variables, performance indexes and constraints. The evolutionary-based approach used to solve the optimization problem is explained in section 4. Numerical simulations of the optimization results and comparisons with no optimized configurations are given in section 5. Finally, some conclusions are drawn in section 6.

2 Knee biped robot dynamics and its transitions

As it was mentioned in the last section, the mathematical model of this system has been explored and developed by several groups. Even though there are still significant differences in the elements of their equations, thus different results were obtained when trying to reproduce them with their values. Therefore, this paper also present the full model to update and clarify those little miscalculations in [11, 12]. In order to keep the relation to previous works and ease of finding the updates, the model is expressed using the same variables.

The continuous dynamic developed by the robot during walking activity is divided into two phases: knee-free phase and knee-locked phase, at the same time knee-strike and heel-strike are the events that produce the transitions in the system. Figure 1 shows a complete schematic diagram of the PBR where: m_h, m_t and m_s are the masses for the hip, thighs and shanks respectively; γ is the angle slope; the variables q_1, q_2 and q_3 are the orientations of each link measured from vertical, they represent the supported leg, free thigh and free shank respectively; remaining variables are the kinematic parameters which meet the next expressions: $a_1+b_1=l_s$, $a_2+b_2=l_t$ and $l_s+l_t=L$.

For the development and simulation of this system, the following statements are considered: all masses are punctual; links representing each leg are identical; transitions are instantaneous, each collision is considered completely inelastic and without sliding.

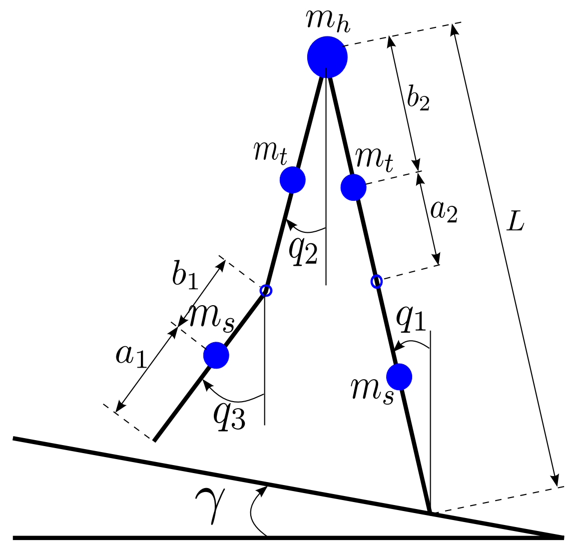


Figure 1. Schematic of the knee biped robot

2.1 Knee-free dynamics phase

This stage will take place after the heel-strike event occurs and will go on until the knee-strike. The whole model is developed using Lagrange formulation. This equation is shown in (1), and it is expressed in the joint-space formulation.

$$\mathbf{H}(\mathbf{q}) + \mathbf{B}(\mathbf{q},\dot{\mathbf{q}}) + \mathbf{G}(\mathbf{q}) = 0 \tag{1}$$

where: $\mathbf{H}(\mathbf{q}) \in \mathbb{R}^{3\times3}$ is the inertia matrix; $\mathbf{B}(\mathbf{q},\dot{\mathbf{q}}) \in \mathbb{R}^{3\times3}$ is the matrix of centrifugal and Coriolis forces; $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^{3\times1}$ is the gravity vector and $\mathbf{q} \in \mathbb{R}^{3\times1}$ is the vector of joint variables, which value is $\mathbf{q} = [q_1 \;\; q_2 \;\; q_3]^T$. The elements of (1) are addressed in (2).

$$\begin{aligned}
H_{11} &= m_s a_1^2 + m_t(l_s+a_2)^2 \\
&\quad +(m_h+m_s+m_t)L^2 \\
H_{12} &= -(m_t b_2 + m_s l_t)L\cos(q_1-q_2) \\
H_{13} &= -m_s b_1 L\cos(q_1-q_3) \\
H_{21} &= H_{12} \\
H_{22} &= m_t b_2^2 + m_s l_t^2 \\
H_{23} &= m_s l_t b_1 \cos(q_2-q_3) \\
H_{31} &= H_{13} \\
H_{32} &= H_{23} \\
H_{33} &= m_s b_1^2 \\
B_{11} &= 0 \\
B_{12} &= -(m_t b_2 + m_s l_t)L\sin(q_1-q_2)\dot{q}_2
\end{aligned}$$

$$
\begin{aligned}
B_{13} &= -m_s L b_1 \sin(q_1 - q_3)\dot{q}_3 \\
B_{21} &= -(m_t b_2 + m_s l_t) L \sin(q_1 - q_2)\dot{q}_1 \\
B_{22} &= 0 \\
B_{23} &= m_s l_t b_1 \sin(q_2 - q_3)\dot{q}_3 \\
B_{31} &= m_s L b_1 \sin(q_1 - q_3)\dot{q}_1 \\
B_{32} &= -m_s l_t b_1 \sin(q_2 - q_3)\dot{q}_2 \\
B_{33} &= 0 \\
G_1 &= -(m_s a_1 + m_t(l_s + a_2) \\
&\quad +(m_h + m_t + m_s)L) g \sin(q_1) \\
G_2 &= (m_t b_2 + m_s l_t) g \sin(q_2) \\
G_3 &= m_s b_1 g \sin(q_3)
\end{aligned} \tag{2}
$$

2.2 Knee-locked dynamics phase

The knee-locked period starts just after the knee-strike event has occurred and it finishes just before the heel-strike. Up to this point, the new system is compared to the Compass Gait model and its dynamic can also be obtained from Lagrangian formulation. In the same way, this equation has the structure of equation (1), but dimension has changed to: $\mathbf{H}(\mathbf{q}) \in \mathbb{R}^{2\times2}$, $\mathbf{B}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{2\times2}$ and $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^{2\times1}$, and its components are described in (3) where $\mathbf{q} = [q_1 \;\; q_2]^T$

$$
\begin{aligned}
H_{11} &= m_s a_1^2 + m_t(l_s + a_2)^2 + (m_h + m_s + m_t)L^2 \\
H_{12} &= -(m_t b_2 + m_s(l_t + b_1)) L \cos(q_1 - q_2) \\
H_{21} &= H_{12} \\
H_{22} &= m_t b_2^2 + m_s(l_t + b_1)^2 \\
B_{11} &= 0 \\
B_{12} &= -(m_t b_2 + m_s(l_t + b_1)) L \sin(q_1 - q_2)\dot{q}_2 \\
B_{21} &= (m_t b_2 + m_s(l_t + b_1)) L \sin(q_1 - q_2)\dot{q}_1 \\
B_{22} &= 0 \\
G_1 &= -(m_s a_1 + m_t(l_s + a_2)) g \sin(q_1) \\
&\quad -(m_h + m_t + m_s) L g \sin(q_1) \\
G_2 &= (m_t b_2 + m_s(l_t + b_1)) g \sin(q_2)
\end{aligned} \tag{3}
$$

2.3 Knee-strike event

Knee-strike is modeled as an event that takes into account the propagation effects of the collision at the swing knee. For one thing the only force actuating in the system is located at the tip of the stance leg because of the friction, and the total angular momentum is conserved around this point. In addition, the angular momentum of the swing thigh and swing shank is conserved around the hip. From this two conditions, the result velocities after the event can be calculated according to (4), and its components are addressed in (6). It is important to point out that the velocity of the swing shank become equal to the swing thigh and it is expressed in (5).

$$
\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}^+ = \left(Q^+\right)^{-1} Q^- \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix}^- \tag{4}
$$

$$
\dot{q}_3^+ = \dot{q}_2^+ \tag{5}
$$

$$
\begin{aligned}
\alpha &= q_1 - q_2 \\
\beta &= q_1 - q_3 \\
\gamma &= q_2 - q_3 \\
Q_{11}^- &= -(m_s l_t + m_t b_2) L \cos\alpha - m_s b_1 L \cos\beta \\
&\quad +(m_t m_s m_h) L^2 + m_s a_1^2 + m_t(l_s + a_2)^2 \\
Q_{12}^- &= -(m_s l_t + m_t b_2) L \cos\alpha + m_s b_1 l_t cos\gamma \\
&\quad +m_t b_2^2 + m_s l_t^2 \\
Q_{13}^- &= -m_s b_1 L \cos\beta + m_s b_1 l_t \cos\gamma + m_s b_1^2 \\
Q_{21}^- &= -(m_s l_t + m_t b_2) L \cos\alpha - m_s b_1 L \cos\beta \\
Q_{22}^- &= m_s b_1 l_t \cos\gamma + m_s l_t^2 + m_t b_2^2 \\
Q_{23}^- &= m_s b_1 l_t \cos\gamma + m_s b_1^2 \\
Q_{11}^+ &= Q_{21}^+ + m_t(l_s + a_2)^2 + m_s a_1^2 \\
&\quad +(m_h + m_t + m_s) L^2 \\
Q_{12}^+ &= Q_{21}^+ + m_s(l_t + b_1)^2 + m_t b_2^2 \\
Q_{21}^+ &= -(m_s(b_1 + l_t) + m_t b_2) L \cos\alpha \\
Q_{22}^+ &= m_s(l_t + b_1)^2 + m_t b_2^2
\end{aligned} \tag{6}
$$

2.4 Heel-strike event

Heel-strike is modeled as an event that considers the propagation effects of the inelastic collision at the tip of the swing leg with the ground. Given that the only force that acts on the system is located at the point of collision, the angular momentum is conserved around this point. Subsequent to this the angular momentum of the new swing leg is also conserved around the hip because there is no any external momentum. From this two conditions, the result velocities after the event can be calculated according to (7), and its components are addressed in (9). It is important to point out that the velocity of the new swing shank remains equal to the new swing thigh until this precise moment, and it is expressed in (8).

$$
\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}^+ = \left(Q^+\right)^{-1} Q^- \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}^- \tag{7}
$$

$$
\dot{q}_3^+ = \dot{q}_2^+ \tag{8}
$$

$$
\begin{aligned}
\alpha &= q_1 - q_2 \\
Q_{11}^- &= Q_{12}^- + (m_h L + 2m_t(a_2 + l_s) \\
&\quad +2m_s a_1)L\cos\alpha \\
Q_{12}^- &= -m_s a_1(l_t + b_1) - m_t b_2(l_s + a_2) \\
Q_{21}^- &= Q_{12}^- \\
Q_{22}^- &= 0 \\
Q_{11}^+ &= Q_{21}^+ + (m_h + m_t + m_s)L^2 \\
&\quad +m_s a_1^2 + m_t(a_2 + l_s)^2 \\
Q_{12}^+ &= Q_{21}^+ + m_s(b_1 + l_t)^2 + m_t b_2^2 \\
Q_{21}^+ &= -(m_s(b_1 + l_t) + m_t b_2)L\cos\alpha \\
Q_{22}^+ &= m_s(l_t + b_1)^2 + m_t b_2^2 \qquad (9)
\end{aligned}
$$

3 Optimization Problem Statement

As it was mentioned in section 1, one of the central requirements for the design of PBR is the fulfillment of a particular job. This one is, for this paper, to take 15 kilograms down a slope with four degrees of inclination. Whose measure, in principle, is infinite, but the experiment can be stopped at any time. Also, the motion of the system must be driven only by the role of the gravity vector.

Generally, the PBR design problem is proposed as an optimization problem, where the design variables, that optimize the performance function proposal, are wanted, subject to constraints inherent to the system. Below are mentioned in detail the design variables, performance function and constraints that are part of the problem of optimal PBR design.

3.1 Design Variables

The overall system performance is linked to the combination of kinematic and dynamic parameters. As proposed by the vector $p \in \mathbb{R}^6$, described in (10), for the synthesis of the system. Where the elements of this vector physically represent the location of the mass center, the dimension of the mass and the total length of the foot, which will have each of the limbs. The chosen parameters, create the set of design variables that can be accessed to modify the system performance. It should be noted that the entire dynamics of the walker depends on the set formed by the union of the parameters in p, and the load, which is represented by the magnitude and location of the mass at the hip, however, this parameter is fixed to fulfill the task.

$$
\begin{aligned}
p &= [x_1, x_2, x_3, x_4, x_5, x_6]^T \\
&= [a_1, b_1, a_2, b_2, m_t, m_s]^T \qquad (10)
\end{aligned}
$$

3.2 Objective function

The conditions of the design problem are reflected as the objectives or performance functions in the optimization problem. For this particular problem, it requires finding a kinematic and dynamic configuration that produces a stable limit cycle with symmetric and periodic gait. Therefore, the objective function is proposed as shown in the expression (11), where $\Phi \in \mathbb{R}^1$. This function represents the deviation of the current from the previous step, in a sample of the last 10 steps. In equation (11), variable θ represents the angle between the supported leg and the swing thigh during each step and T represents the time of one step performed by one of the legs.

$$\Phi = \int_0^{10T} [\theta(t - T) - \theta(t)]^2 dt \qquad (11)$$

3.3 Constraints

The constraints imposed to the optimization problem are the maximum and minimum lengths for the links and their mass dimension. These restrictions are expressed in mathematical form in (12)-(13). Likewise, this problem is subjected to dynamic equations of the system and its transitions, meaning equations (1)-(9).

$$0 < \quad x_i \quad \leq 0.5[m] \qquad i = 1,2,3,4 \qquad (12)$$
$$0 < \quad x_{i+4} \quad \leq 10[kg] \qquad i = 1,2 \qquad (13)$$

3.4 Additional Considerations for this Particular System

It has been shown in some studies, that the region of attraction for this system to reach and remain in a stable limit cycle is considerably short [13], [14]. For this reason it was used a further optimization problem to provide each individual in the population [7] with the initial conditions within the limit cycle. This was done in order to evaluate correctly each individual in the population and thus to exploit the performance of each one of them. For this lower-level problem, the same objective function as the main optimization problem has been proposed. The result of this process is the set of initial conditions for each individual.

3.5 PBR design problem as an optimization problem

Finally, the PBR design problem is formulated as an optimization problem in order to obtain the vector $p^* \in \mathbb{R}^6$ that optimizes $\Phi \in \mathbb{R}^1$ in (14), subject to (15)-(18), it means:

$$\Phi(\bar{p}^*) = min[\Phi_1] \qquad (14)$$

Subject to:

$$g_i = -x_i < 0 \qquad i = 1, \ldots, 4 \qquad (15)$$
$$g_{i+4} = x_i - 0.5 \leq 0 \qquad i = 1, \ldots, 4 \qquad (16)$$
$$g_{i+8} = -x_{i+4} < 0 \qquad i = 1, \ldots, 2 \qquad (17)$$

$$g_{i+10} = x_{i+4} \leq 10 \qquad i = 1, \ldots, 2 \tag{18}$$

As well as the dynamic equations of the system and its transitions, meaning equations (1)-(9).

4 Differential Evolution Algorithm

The Differential Evolution (DE) algorithm is part of the evolutionary algorithms, which base their operation on the process of natural selection proposed by C. Darwin. These listed as heuristic algorithms, do not guarantee obtaining a global optimum, however, produce a satisfactory solution in reasonable computation time [7].

Some of the benefits of the EA are: 1) as such techniques are based on population, a minimum solution can be reached, but not for all the problems, 2) the start of the search for the optimal solution does not need additional information, that is, it is not necessary the calculation of the gradients, Hessian matrix or proposal of a initial point, to mention a few, 3) this methodology allows to attack complex problems such as non-linear and discontinuous models with minimal changes compared to the gradient methods.

The implemented algorithm for solving the PBR design problem is shown in Table 1.

Table 1. Used algorithm

$G = 0$
Creat a random initial population $\vec{x}^i_G \forall i, i = 1, \ldots, NP$
Find the initial conditions for each individual
Evaluate $\vec{x}^i_G$for each individual
and every constraint $\forall i, i = 1, \ldots, NP$
    For $G = 1$ to MAX_GENERATIONS Do
        for $i = 1$ to NP Do
        Select randomly $r_1 \neq r_2 \neq r_3$:
        $j_{rand} = randint1, D$
        for $j = 1$ to D Do
            if($rand_j[01] < CR$ or $j = j_{rand}$) Then
                $u^i_{j,G+1} = x^{r3}_{j,G} + F(x^{r1}_{j,G} - x^{r2}_{j,G})$
            Else
                $u^i_{j,G+1} = x^i_{j,G}$
            End if
        End for
        Find the initial conditions for each individual
        Evaluate u_{G+1} for each function and every constraint
        If (u_{G+1} is better than $\vec{x^i_{j,G}}$
            $\vec{x}^i_{G+1} = \vec{u}^i_{G+1}$
        Else
            $\vec{x}^i_{G+1} = \vec{x}^i_G$
        End if
        End for
    $G = G + 1$
    End for
End

5 Results

The parameters used in the optimization algorithm are shown in Table 2. As stated in section 3, the results of the optimization process, correspond to the location of the center of mass, the mass dimension for tips and foot length. For assessing equation (11), the dynamics in (1) was numerically integrated, taking into account the conditions for switching from one foot to the other. And the variables of the expression (14) are only stored in an order step by step, for further evaluation. The optimal parameters obtained as solution to this optimization process are shown in (19), which turned out to be the best individual of the generation number 100 in assesing (14). The initial conditions for starting locomotion are written in equation (20).

number of generations	100
number of individuals	50
crosse factor	0.7
step of integration for ODE	0.001

Table 2. Algorithm parameters

$$\begin{aligned}
p_1^* &= 0.3095 \quad [m] \\
p_2^* &= 0.0212 \quad [m] \\
p_3^* &= 0.0045 \quad [m] \\
p_4^* &= 0.0431 \quad [m] \\
p_5^* &= 7.6649 \quad [Kg] \\
p_6^* &= 5.3501 \quad [Kg]
\end{aligned} \tag{19}$$

$$\begin{aligned}
q_1 &= 0 \quad [rad] \\
q_2 &= 0 \quad [rad] \\
q_3 &= 0 \quad [rad] \\
\dot{q_1} &= -0.9469 \quad [rad/s] \\
\dot{q_2} &= 2.0047 \quad [rad/s] \\
\dot{q_3} &= 7.5168 \quad [rad/s]
\end{aligned} \tag{20}$$

5.1 System performance with obtained parameters and comparison

In order to describe the value of the method proposed in this work for the parameters selection, a comparison is made between the parameters obtained and some other sets. But not all the sets of parameters for the system in combination with the initial conditions allow the study of their behavior, because many of them cause the walker to fall. For this purpose, an optimization process has been made in search of initial conditions that allow the study of that configuration.

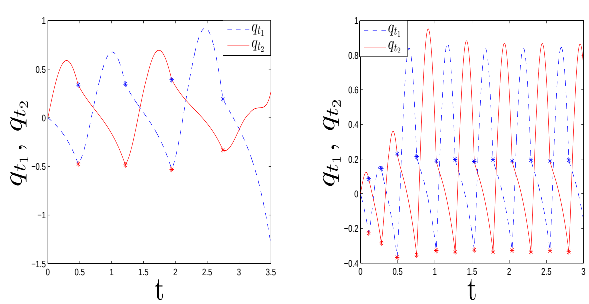
In the subsequent paragraphs, the Figures with the results are going to be described. This pictures only contain the information of two variables: the shank and thigh supported, it means when they remain together and can be described by q_1; and the free thigh which is fully described by q_2.

The first set of parameters, to be compared, are the same shown in [11], where the parameters m_h and γ have been changed for the requirements of the job and the proper initial conditions for that configuration are expressed in (21). The response of this configuration is shown in Figures 2(a)-4(a). It is essential to observe that, at the very beginning; it seems to fulfill the task, however, at the end it falls.

Otherwise, the second set of parameters, to be compared, is a variation of the solution obtained in this document. The resulting solution vector in (19) is replaced, adding a perturbation on the first component. The first element of the solution vector is multiplied by 0.8, and the proper initial conditions for that configuration are expressed in (22), after getting them. The response of this configuration is shown in Figures 2(b)-4(b).

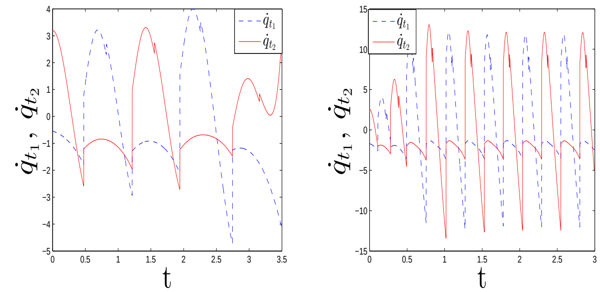
$$\begin{aligned} q_1 &= 0 \quad [rad] \\ q_2 &= 0 \quad [rad] \\ q_3 &= 0 \quad [rad] \\ \dot{q}_1 &= -0.5467 \quad [rad/s] \\ \dot{q}_2 &= 3.1706 \quad [rad/s] \\ \dot{q}_3 &= 3.6471 \quad [rad/s] \end{aligned} \tag{21}$$

$$\begin{aligned} q_1 &= 0 \quad [rad] \\ q_2 &= 0 \quad [rad] \\ q_3 &= 0 \quad [rad] \\ \dot{q}_1 &= -1.7125 \quad [rad/s] \\ \dot{q}_2 &= 2.2619 \quad [rad/s] \\ \dot{q}_3 &= 3.5530 \quad [rad/s] \end{aligned} \tag{22}$$



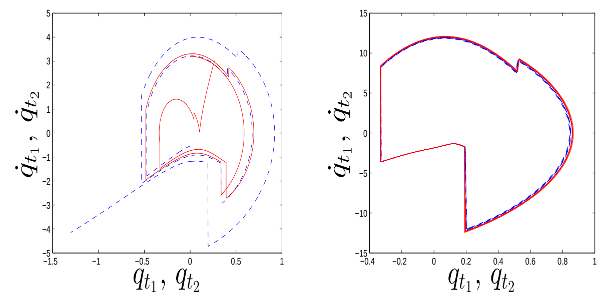
(a) Configuration in [11] (b) Parameter perturbation

Figure 2. Angular position for each leg



(a) Configuration in [11] (b) Parameter perturbation

Figure 3. Angular velocity for each leg



(a) Configuration in [11] (b) Parameter perturbation

Figure 4. Phase portrait

To complete the comparison, Figures 5-7 show the performance of the vector solution obtained in this work. Figures 5 and 6 illustrate how the biped robot begins to walk with short steps and quickly reaches its limit cycle, where it made strides compared with the initial steps. In Figure 7 the limit cycle in which it keeps its symmetric walk is illustrated.

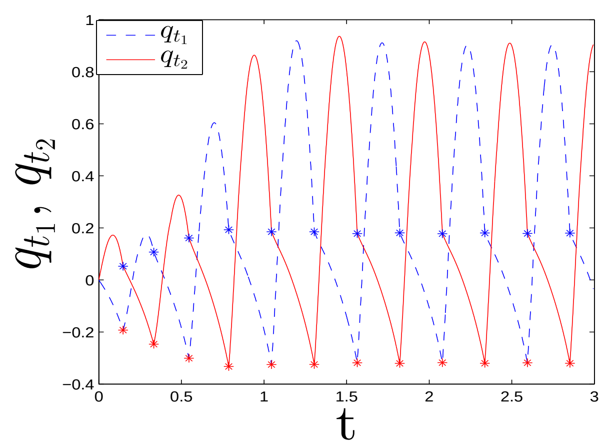


Figure 5. Angular position for each leg of p^*

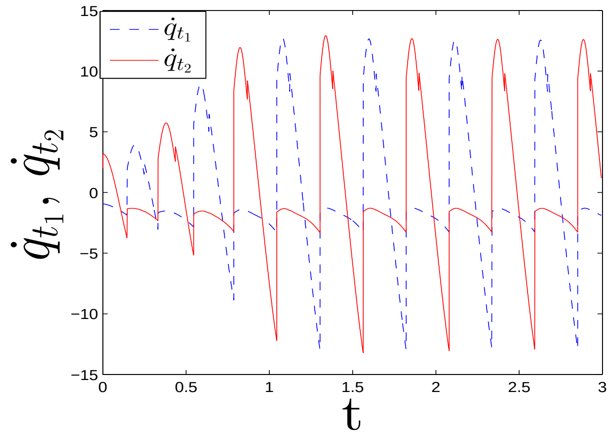


Figure 6. Angular velocity for each leg of p^*

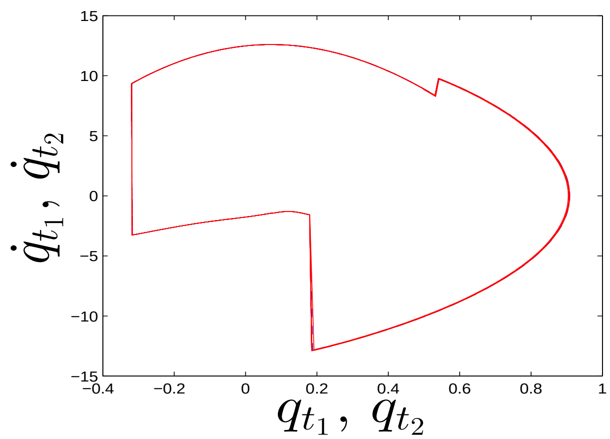


Figure 7. Phase portrait for p^*

Finally, Table 3 shows the numeric value after the evaluation of the objective function in (11) for each vector. Where Φ_1 and Φ_2 represent the first and second set of parameters proposed respectively. And Φ^* represents the parameters obtained in this work. It is important to mention that the Φ_1 value only considers the evaluation of the first two steps given by the biped, since it falls down. This results prove that the solution obtained finishes the work proposed without any problem, gracefully and with a desired behavior.

Φ_1	Φ_2	Φ^*
34.3349	1.7206	0.0043

Table 3. Objective function evaluation

6 Conclusions

In this paper, a successful solution to the design problem of a kneed bipedal robot configuration has been developed. With established conditions and thus fulfilling a particular job. Allowing to explain the importance of obtaining the parameters to show the potential applications of this system beyond just walking.

This research will affect positively the way to conceive building dynamic biped robots and put them together with some controllers in order to obtain an efficient and capable system, when performing a semi-actuated version or fully actuated. And then take advantage of multiple and powerful controllers that have been already developed for this type of devices.

References

[1] M. Vukobratovich and D. Stokich, Dynamic control of unstable locomotion robots, *Mathematical Biosciences*, 24(12), 1975, 129-157.

[2] S. Mochon and T. A. McMahon, Ballistic walking: an improved model, *Mathematical Biosciences*, 52(3-4), 1980, 241-260.

[3] A. Goswami, B. Espiau and A. Keramane, Limit cycles and their stability in a passive bipedal gait, *Proc. 1996 IEEE International Conference on Robotics and Automation* , Minesota, USA, 1996, 246-251.

[4] M. J. Coleman, M. Garcia, K. Mombaur and A. Ruina, Prediction of stable walking for a toy that cannot stand, *Physical Review E.*, 64(2), 2001, 1-3.

[5] R. Tedrake, T. W. Zhang, M. F. Fong and H. S. Seung, Actuating a simple 3d passive dynamic walker, *Proc. 2004 IEEE International Conference on Robotics and Automation*, Luisiana, USA, 2004, 4656-4661.

[6] E. A. Portilla-Flores, C. Morales-Cruz, R. A Suarez-Santillan, M. G. Villarreal-Cervantes, P. A. Niño-Suarez, M. B. Calva-Yañez, Optimal synthesis of the continuously variable transmission input mechanism using an evolutionary approach, *Int. J. of Innovative Computing Information and Control* , 9(5), 2013, 1851-1864.

[7] A. P. Engelbrecht, *Computational intelligence: an introduction*, (New York: John Wiley and Sons Inc., 2007)

[8] J. E. Pratt and G. A. Pratt, Exploiting natural dynamics in the control of a 3d bipedal walking simulation, *Proc. of International Conference on Climbing and Walking Robots*, Portsmouth, England, 1999.

[9] F. Asano and M. Yamakita, Virtual gravity and coupling control for robotics gait synthesis, *IEEE Transactions on Systems, Man and Cybernetics, Part A: Systems and Humans*, 31(6), 2001, 737-745.

[10] S. H. Collins, M. Wisse and A. Ruina, A three-dimensional passive-dynamic walking robot with two legs and knees, *I. J. Robotic Res.*, 20(7), 2001, 607-615.

[11] Z. Liu, A. Zhang, Y. Tian, P. Zhang and D. Gao, Further analysis of the kneed passive-dynamic biped robot, *Control and Decision Conference*, 2010, 1777-1782.

[12] V. F. Hsu Chen, *Passive Dynamic Walking with knees: A Point Foot Model*, (Massachusetts Institute of Technology 2007).

[13] M. W. Spong, Passivity based control of the compass gait biped, *Proc. of IFAC World Congress*, Beijing, China, 1999,

[14] I. R. Manchester, M. M. Tobenkin, M. Levashov and R. Tedrake, Regions of attraction for hybrid limit cycles of walking robots, *Proc. of IFAC World Congress*, Milano, Italy, 2011, 5801-5806.